<u>Assignment 1 (Weeks 1 and 2)</u>

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Part I: Short Answer Questions

Q1. Difference : A file system stores data as raw files with no centralized control, while a DBMS manages data through a structured layer that handles access, integrity, and consistency automatically.

Scenario: Consider an e-commerce platform like an online store where the warehouse team, the customer support team, and the payments team all maintain their own separate spreadsheets for orders. A customer cancels an order — the payments team updates their sheet and processes the refund, but the warehouse team never sees the update and ships the product anyway. With a DBMS, all three teams query the same central orders table in real time, so the cancellation is immediately visible to everyone and the shipment is automatically blocked.

Q2. A strong entity has its own unique identifier and can exist independently in the database. A weak entity has no primary key of its own and depends on a parent (owner) strong entity to exist.

The primary key of a weak entity is formed as a Composite Key by combining the *Primary Key of its parent strong entity* with the weak entity's own *Partial Key* (discriminator). The association connecting them is strictly called an Identifying Relationship, represented by a double diamond in an ER diagram.

Q3. Phone number should be modelled as a MULTIVALUED ATTRIBUTE because a single student can have more than one phone number (personal, home, etc.). A simple attribute holds exactly one value, and a composite attribute is one that can be broken into sub-parts like first and last name. Since the issue here is multiple values of the same kind, multivalued is the correct choice. It is represented using a double oval in an ER diagram.

Q4. Difference: A SUPER KEY is any single attribute or set of attributes that can uniquely identify a row within a table, regardless of whether it contains extra, unnecessary columns. A CANDIDATE KEY is a *minimal* super key; it contains no redundant attributes, meaning if you remove even one column from it, it loses its uniqueness property.

Candidate Keys: Assuming both institutional roll numbers and personal emails are unique, the candidate keys are:

- ➔ {roll no}
- ➔ {email}

Super keys that are NOT candidate keys:

- ➔ {roll no, name} — redundant because roll no alone is enough
- ➔ {roll no, email, branch} — redundant for the same reason

Q5.

- The constraint violated is the **Referential Integrity Constraint**
- It protects against orphan records - rows in a child table that point to a parent record that doesn't exist, which would break the logical relationship between tables
- The insert operation should be rejected by the DBMS

Q6.

- A separate **JUNCTION/BRIDGE TABLE** must be created to resolve the M:N relationship
- Its primary key is a composite key made up of the primary keys of both participating entity tables
- The relationship cannot be represented as a column in one table because a single column can only hold one value. Storing multiple foreign key references inside a single column would violate the domain constraint, which requires every value to be atomic and indivisible.

Part II: Long Answer Question

Part A – Entity and Relationship Identification

Entities and key attributes:

- Customer → CustomerID, Name, Phone, Address
- Restaurant → RestaurantID, Name, Address, Phone
- MenuItem → MenuItemID, Name, Price
- Order → OrderID, OrderDate, Status
- **Relationships:**

Relationship	Cardinality	Justification
Customer - Order	1:N	One customer can place many orders but each order belongs to exactly one customer
Order - Restaurant	N:1	Each order is placed at exactly one restaurant, but a restaurant can receive many orders

Relationship	Cardinality	Justification
Order - MenuItem	M:N	One order can contain multiple menu items and the same menu item can appear in many orders
Restaurant - MenuItem	1:N	One restaurant offers many menu items but each menu item belongs to one restaurant

Part B – Attribute Classification

Multivalued Attribute: Restaurant_PhoneNumbers. This qualifies because a business restaurant often maintains multiple active communication lines (e.g., landline, reception mobile, and manager's line) linked to a single restaurant profile.

Composite Attribute: Customer_Address. This qualifies because a delivery address is not a single indivisible unit, it is naturally broken down into sub-attributes like Street Address, City, State, and Zip Code for delivery mapping.

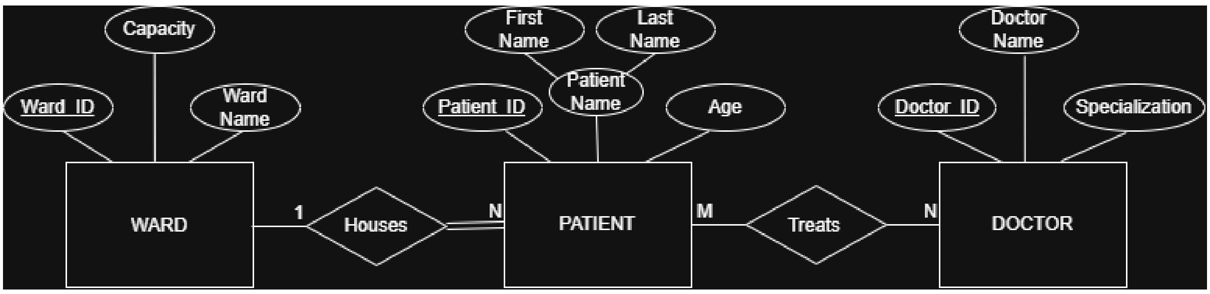
Part C – Relational Schema Design

Order_Details(OrderID, MenuItemID,Quantity)

- **Primary key:** (OrderID, MenuItemID) - the combination uniquely identifies each line item within an order.
- **Foreign Keys:**
  - OrderID references the primary key of the Order table.
  - MenuItemID references the primary key of the MenuItem table.
- **Additional attribute:** Quantity - it stores how many units of that menu item were ordered. Price – it stores price of the menu item.

Part III: ER Diagram Design :

Q1.



Q2.

The Doctor–Patient relationship is M:N because in a hospital setting, a patient can be treated by multiple doctors simultaneously and each doctor manages multiple patients at the same time. This creates an associative entity called Treatment, whose composite primary key would consist of {DoctorID, PatientID}.

For participation constraints: the Patient–Ward relationship has total participation on the Patient side because every patient recorded in the system must be physically admitted to a ward, a patient cannot exist in the hospital database without being assigned to one. The Ward side has partial participation since a ward can exist in the system before any patient is admitted to it. For the Doctor–Treats–Patient relationship, both sides have partial participation, a doctor may be registered in the system before being assigned any patient, and similarly a patient might exist briefly before a doctor is assigned.

Finally, a key attribute design choice was breaking down the patient's name into a composite attribute containing *First Name* and *Last Name*. This design allows the hospital administrative staff to sort records alphabetically and easily filter patient lookups by family names.